COLUMBIA UNIVERSITY

ROBOTICS STUDIO MECE4611 SPRING 2025
ASSIGNMENT 4

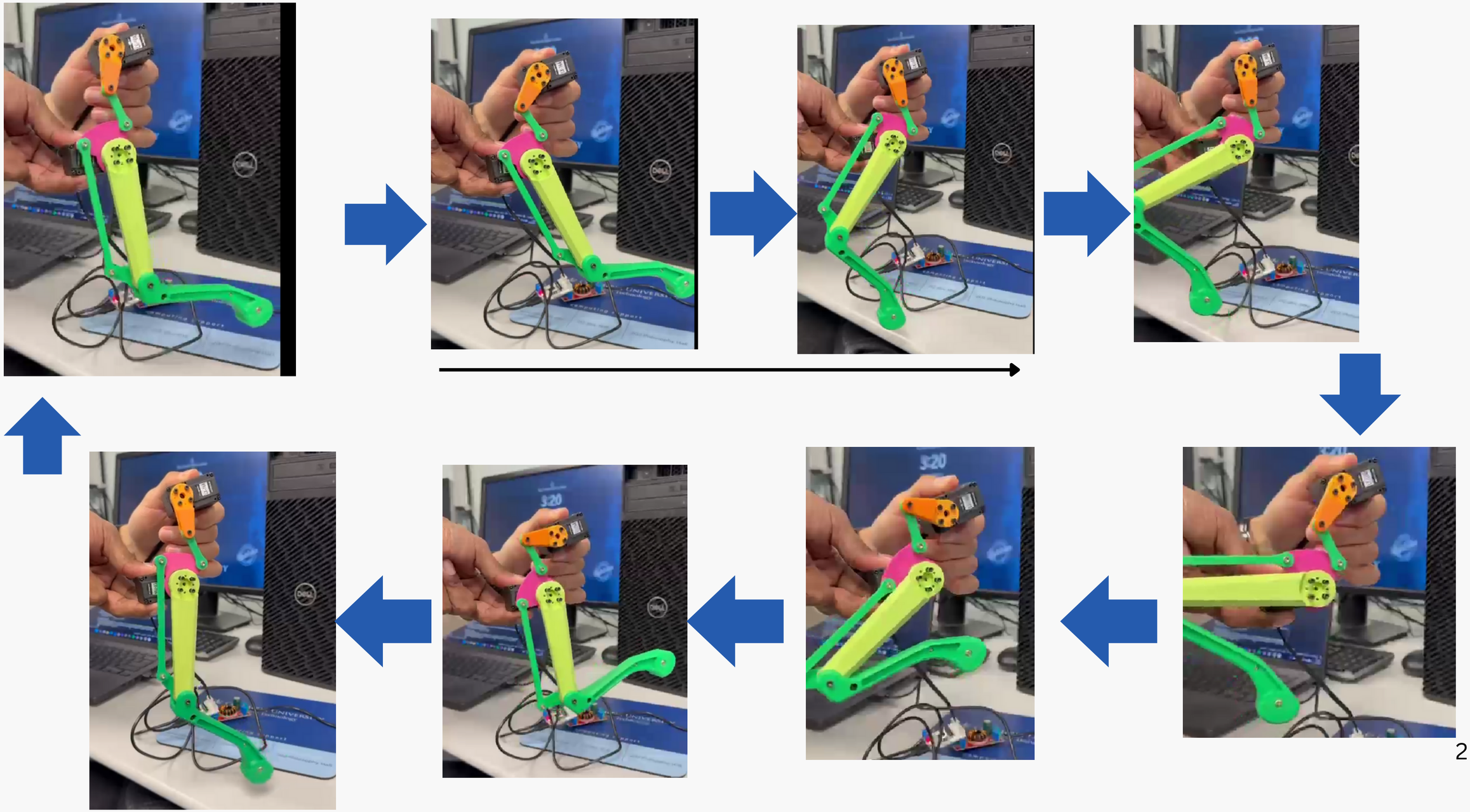
1.YASH MANGROLE(ym3067)
2.ABHISHEK RASAL(ar4893)

Kruger (Quadrapedal Robot)



Date - 03/02/2025

Leg Motion of Kruger

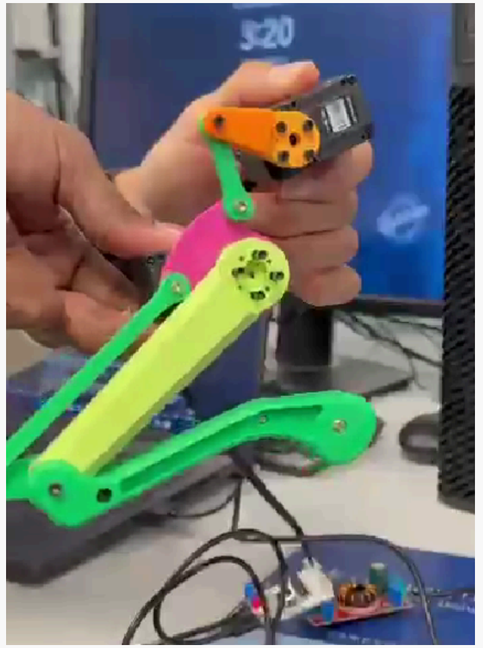


Leg Motion of Kruger Video

https://drive.google.com/file/d/1w5T3N-5B1o2JqB9dnsWIKcZQCd3il7GZ/view?usp=drive_link

Extreme Leg Position of Kruger





Code for Execution

C: > Users > yashm > OneDrive > Desktop > SEM2 > robotics Studio > leg_move.py > ...

```python
#!/usr/bin/env python3
from math import sin, cos, pi
from pylx16a.lx16a import *
import time

# Constants for the sinusoidal motion
HIP_SERVO_ID = 5       # ID of the hip servo
KNEE_SERVO_ID = 8      # ID of the knee servo
AMPLITUDE = 60         # Amplitude of motion in degrees
CENTER_POS = 120       # Center position for servos (degrees)
FREQUENCY = 0.1        # Controls the speed of the motion
PHASE_DIFF = pi/2      # Phase difference between hip and knee (90 degrees)

def main():
    # Initialize the bus - change this to your actual port
    # Windows: "COM3" or similar
    # Linux: "/dev/ttyUSB0"
    # Mac: "/dev/tty.usbserial-*"
    PORT = "COM6"  # Change to your actual port

    try:
        # Initialize the communication with the servos
        LX16A.initialize(PORT)

        # Initialize servo objects
        hip_servo = LX16A(HIP_SERVO_ID)
        knee_servo = LX16A(KNEE_SERVO_ID)

        # Set angle limits (0-240 degrees is the standard range for LX16A servos)
        hip_servo.set_angle_limits(0, 240)
        knee_servo.set_angle_limits(0, 240)

        print("Servos initialized successfully!")

        # Start sinusoidal motion
        t = 0
        try:
            while True:
                # Calculate servo positions using sine waves with phase difference
                hip_pos = sin(t) * AMPLITUDE + CENTER_POS
                knee_pos = sin(t + PHASE_DIFF) * AMPLITUDE + CENTER_POS

                # Move servos to calculated positions
                hip_servo.move(hip_pos)
                knee_servo.move(knee_pos)

                # Small delay to control the speed
                time.sleep(0.05)
                t += FREQUENCY

        except KeyboardInterrupt:
            print("\nMotion stopped by user")
            # Return servos to center position
            hip_servo.move(CENTER_POS)
            knee_servo.move(CENTER_POS)

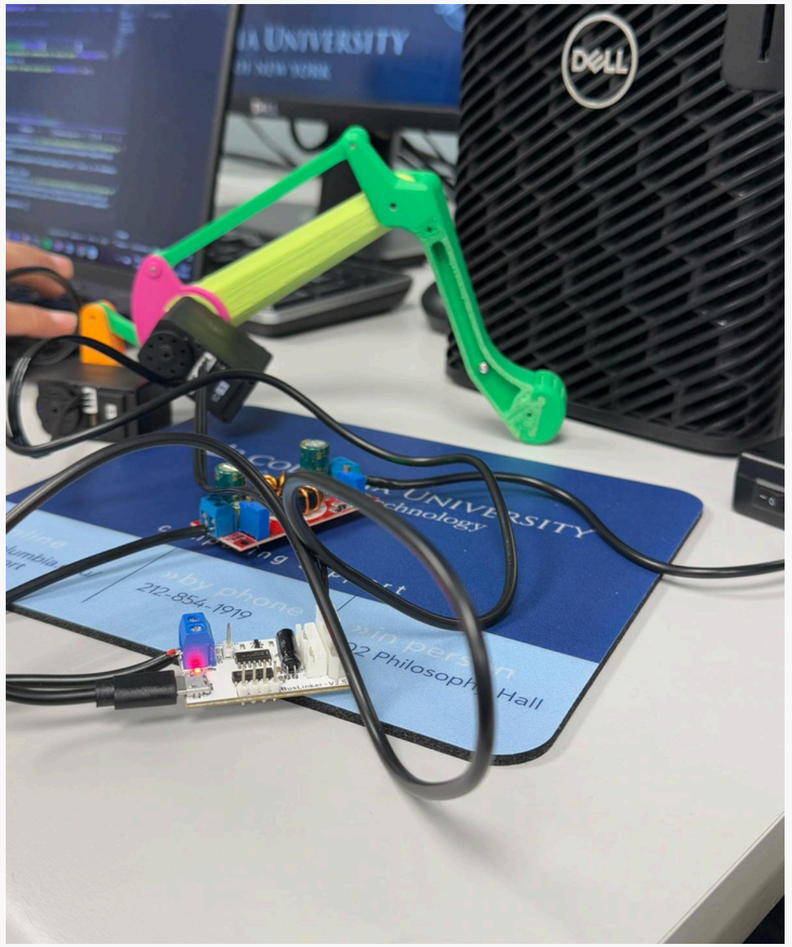
    except ServoTimeoutError as e:
        print(f"Servo {e.id_} is not responding. Exiting...")
    except Exception as e:
        print(f"Error: {e}")

if __name__ == "__main__":
    main()
```

3D-print quality, support structure removed



Cable Routing



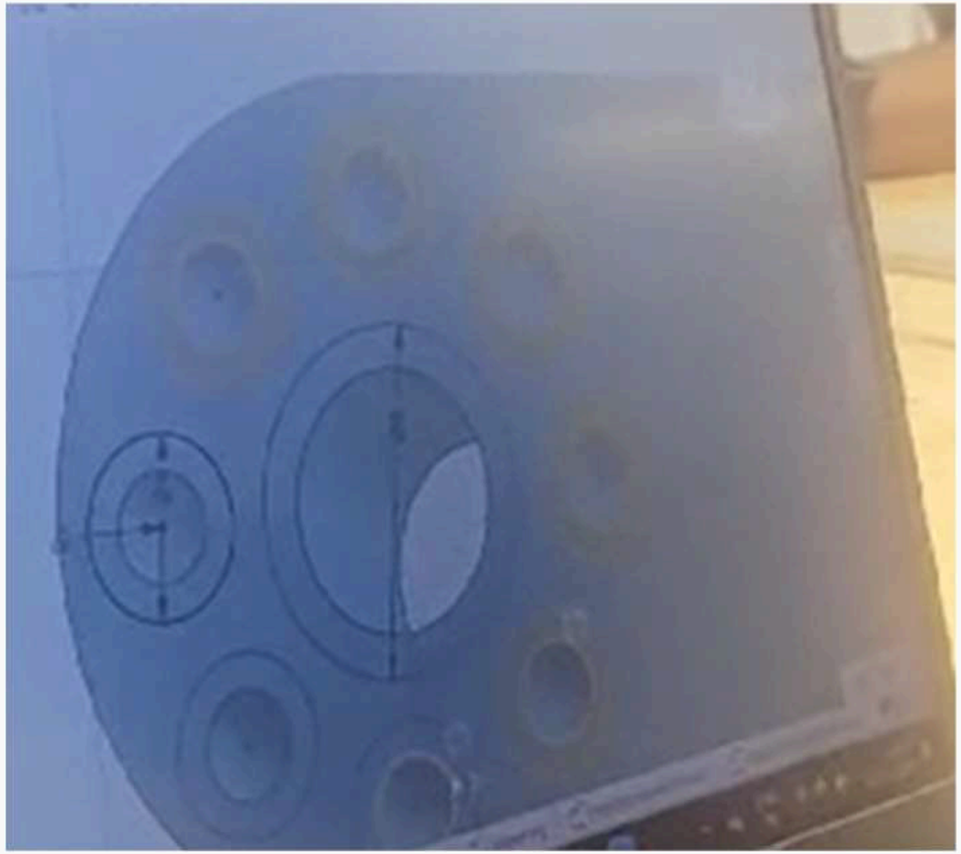
Nuts and Bolts Integration to Kruger





Form and Fit Issues identified

1. Servo motors Shaft was 4 x mm in depth. However, in Grabcad, the models were 3.45mm in depth.
2. The screw holes are smaller when 3d printed so we need to increase the size
3. Leg to servo attachments are too thin, we need to thicken them and apply a counter bore



Posted On Ed

Kruger's Leg motion #136

Yash Vinodsingh Mangrole
7 hours ago in **General**

STAR WATCHING 7 VIEWS

https://drive.google.com/file/d/1w5T3N-5B1o2JqB9dnsWIKcZQCd3il7GZ/view?usp=drive_link

1 Comment Edit Delete •••

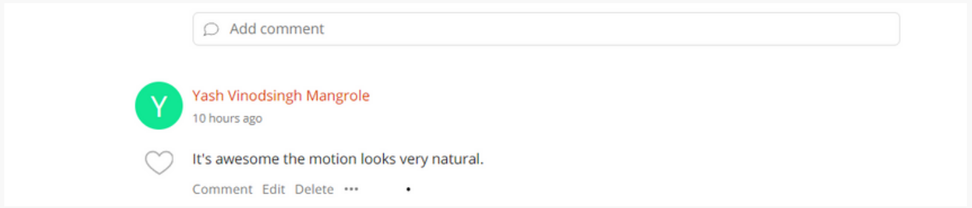
1 Answer

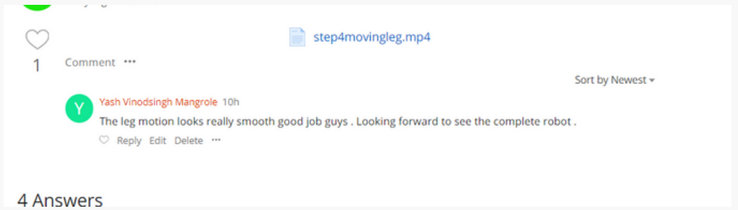
Grace Munger
3 hours ago

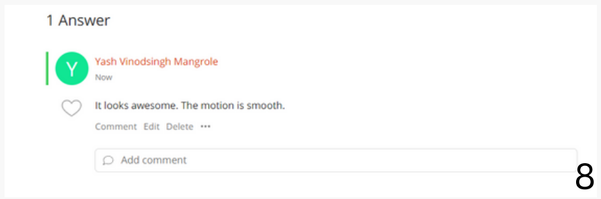
This is a very cool linkage system and I like all of the different colors. Very interested to see this robot walking!

Comment •••

Add comment







Rubrics Attempted

- Title slide complete
- overall aesthetics, layout and formatting of the slides
- Sequence of photos showing leg in motion
- posting video of moving leg on the discussion board at least 24h in advance of deadline, and commenting constructively and positively on at least three other's postings (show screenshots)
- extreme leg positions tested and measured
- form/fit issues identified, listed and addressed (show how)
- all components properly bolted and connected (with inserts)
- 3D-print quality, support structure removed
- Different leg motion patterns explored
- Leg Modularity demonstrated
- Cables routed properly and securely
- Exception handling in code catches motor disconnect

Thank You